

Bayesian Conjugate Priors — Detailed Reference Guide

1. Core Bayesian Pattern

Posterior $\propto$ Likelihood $\times$ Prior

Given data x and parameter θ :

- Prior $\pi(\theta)$
- Likelihood $L(\theta) = P(x | \theta)$
- Posterior $\pi(\theta | x) \propto \pi(\theta)L(\theta)$

2. Conjugate Prior Families

Binomial/Bernoulli:

Prior: Beta(α , β)

Data: S successes out of n

Posterior: Beta($\alpha + S$, $\beta + n - S$)

MAP = $(\alpha + S - 1) / (\alpha + \beta + n - 2)$

Mean = $(\alpha + S) / (\alpha + \beta + n)$

Poisson:

Prior: Gamma(α , β) (rate parameterization)

Data: Sum S over n observations

Posterior: Gamma($\alpha + S$, $\beta + n$)

MAP = $(\alpha + S - 1) / (\beta + n)$

Mean = $(\alpha + S) / (\beta + n)$

Normal (mean known variance):

Prior: Normal(μ_0 , τ_0^2)

Data: n samples with mean $\bar{x}$

Posterior:

Precision: $1/\tau n^2 = 1/\tau 0^2 + n/\sigma^2$

Posterior mean: $\mu n = \tau n^2(\mu 0/\tau 0^2 + n\bar{X}/\sigma^2)$

Multinomial:

Prior: Dirichlet($\alpha_1, \dots, \alpha_K$)

Data: Counts $n_1, \dots, n_K$

Posterior: Dirichlet($\alpha_1 + n_1, \dots, \alpha_K + n_K$)

3. Posterior Predictive Distributions

Beta-Binomial:

$P(X = k \mid \text{data}) = C(m, k) * \text{Beta}(\alpha + S + k, \beta + n - S + m - k) / \text{Beta}(\alpha + S, \beta + n - S)$

Gamma-Poisson:

Predictive distribution is Negative Binomial.

Dirichlet-Multinomial:

Predictive probability for category j is $(\alpha_j + n_j) / (\sum(\alpha_i + n_i))$.

4. General Derivation Template (Reusable)

Step 1: Write the prior kernel.

Step 2: Write the likelihood kernel.

Step 3: Multiply them.

Step 4: Combine exponents/powers.

Step 5: Recognize resulting kernel as known family.

Step 6: Read off updated parameters.

5. Example: Poisson-Gamma

Prior: $\lambda \sim \text{Gamma}(2, 1)$

Likelihood for data $x_1=2, x_2=3, x_3=4$:

$$L(\lambda) \propto \lambda^9 e^{-3\lambda}$$

Posterior:

$$\text{Gamma}(2+9, 1+3) = \text{Gamma}(11, 4)$$

$$\text{MAP} = (11-1)/4 = 2.5$$

$$\text{CE} = 11/4 = 2.75$$

6. Practical Tips

- Always match parameterization (rate vs scale).
- Identify sufficient statistics (S, n) immediately.
- If likelihood monotone in θ , MAP may lie on boundary.
- Conjugacy ensures closed-form posteriors; otherwise use MCMC.